

Lab - Azure SQL Database

To Begin with the Lab

Summary

This lab demonstrates how to create an Azure SQL Database using the Azure Portal. You will create a SQL Server, configure authentication and networking, create a database, connect to it using a SQL client or Azure Query Editor, verify the database is working, and finally delete the resources to avoid unnecessary charges.

Understanding

Azure SQL Database is a fully managed relational database service (Database as a Service - DBaaS) provided by Microsoft Azure. It is based on the Microsoft SQL Server database engine but eliminates the need to manage hardware, operating systems, backups, patching, and high availability. Azure automatically handles updates, security, backups, and scaling, allowing developers to focus on building applications instead of managing infrastructure.

Example:

Suppose you are developing an employee management application. Instead of installing Microsoft SQL Server on a virtual machine, you can create an Azure SQL Database. Your application connects to the database using a connection string, while Azure manages backups, security, and performance automatically.

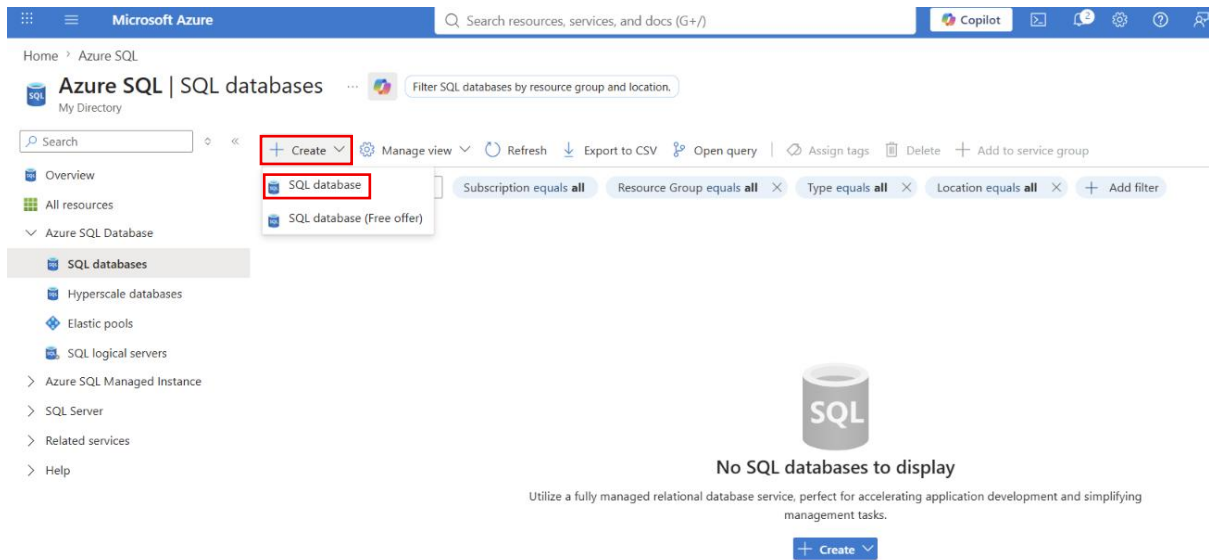
Lab steps

Step 1: Sign in to Azure Portal

- Open the Azure Portal.
 - Sign in using your Azure account.
-

Step 2: Open Azure SQL

- In the search bar, search for **SQL Databases**.
- Open the **SQL Databases** service.
- Click **Create**.
- Select **SQL Database**.



Step 3: Configure Basic Details

- Choose your Azure Subscription.
- Create or select a Resource Group.
- Enter a Database name.
- Create a new SQL Server or select an existing one.

The screenshot shows the 'Create SQL Database' form in the Microsoft Azure portal. The form is divided into two main sections: 'Project details' and 'Database details'. In the 'Project details' section, there are two dropdown menus: 'Subscription *' with the value 'Visual Studio Enterprise Subscription' and 'Resource group *' with the value 'terraform-storage-rg'. Below the resource group dropdown is a link 'Create new'. In the 'Database details' section, there are two more dropdown menus: 'Database name *' with the value 'SQLsvr-dev-eus' and 'Server *' with the value 'Select a server'. Below the server dropdown is a link 'Create new'. The form is styled with a light blue header and a white body. The 'Project details' section has a subtitle 'Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.' The 'Database details' section has a subtitle 'Enter required settings for this database, including picking a logical server and configuring the compute and storage resources'.

Step 4: Configure SQL Server

- Enter a **unique** Server name.
- Choose the **Region**.
- Select **SQL Authentication**.
- Enter the **Administrator username**.
- Enter and confirm the **Administrator password**.
- Click **OK** or **Review** to continue.

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

Home > Azure SQL | SQL databases > Create SQL Database

Create SQL Database Server

Microsoft

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name * web-db ✓

Location * (Asia Pacific) East Asia ✓

Authentication

Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. We recommend using only Microsoft Entra authentication [Learn more](#) using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#)

Authentication method

☐ Use Microsoft Entra-only authentication

☐ Use both SQL and Microsoft Entra authentication

☒ Use SQL authentication

Server admin login * SQLAdmin ✓

Password * ***** ✓

Confirm password * ***** ✓

OK

Step 5: Configure Compute and Storage

- Select the desired Workload type **development**.
- Click on the **Configure database**.

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Home > Azure SQL | SQL databases

Create SQL Database

Microsoft

Database name * SQLSERVER-DEV-TEST ✓

Server * (new) web-db (East Asia) ✓

Want to use SQL elastic pool? ☐ Yes ☒ No

Workload environment

☒ Development

☐ Production

Default settings provided for Development workloads. Configurations can be modified as needed.

Compute + storage * ☐ General Purpose - Serverless

Standard-series (Gen5), 1 vCore, 32 GB storage, zone redundant disabled

[Configure database](#)

Backup storage redundancy

Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

Backup storage redundancy ☐ Locally-redundant backup storage

☐ Zone-redundant backup storage

☒ Geo-redundant backup storage

☐ Geo-Zone-redundant backup storage

- Choose the Compute tier **provisioned**.

Configure



Service and compute tier

Select from the available tiers based on the needs of your workload. The vCore model provides a wide range of configuration controls and offers Hyperscale and Serverless to automatically scale your database based on your workload needs. Alternately, the DTU model provides set price/performance packages to choose from for easy configuration. [Learn more](#)

SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Service tier

General Purpose (Most budget friendly)

[Compare service tiers](#)

Compute tier

☒ **Provisioned** Compute resources are pre-allocated. Billed per hour based on vCores configured.

☐ **Serverless** - Compute resources are auto-scaled. Billed per second based on vCores used.

- Choose Service tier **Basic** (For less demanding workload).

Configure



Service and compute tier

Select from the available tiers based on the needs of your workload. The vCore model provides a wide range of configuration controls and offers Hyperscale and Serverless to automatically scale your database based on your workload needs. Alternately, the DTU model provides set price/performance packages to choose from for easy configuration. [Learn more](#)

SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Service tier

General Purpose (Most budget friendly)

Compute tier

vCore-based purchasing model

General Purpose (Most budget friendly)

Hyperscale (Highly scalable compute and storage)

Business Critical (Highest availability and performance)

DTU-based purchasing model

Basic (For less demanding workloads)

Compute Hardware

Select the hardware configuration based on your workload. Confidential computing hardware depends on your workload.

Standard (Budget friendly)

Hardware Configuration

Premium (Highest availability and performance)



Cost summary

General Purpose (GP, Gen5, 2)

Cost per vCore (in INR) ¹ 13774.23

vCores selected x 2

Cost per GB (in INR) 14.35

Max storage selected (in GB) x 41.6

ESTIMATED COST / MONTH 28145.46 INR

NOTES

¹ The dev/test discount has been automatically applied for your selected subscription. [Learn more](#)

- Select the required **Storage size**.
- Click on the **Apply**.

Configure ...

 Feedback

Service and compute tier

Select from the available tiers based on the needs of your workload. The vCore model provides a wide range of configuration controls and offers Hyperscale and Serverless to automatically scale your database based on your workload needs. Alternately, the DTU model provides set price/performance packages to choose from for easy configuration. [Learn more](#)

 SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Service tier

Basic (For less demanding workloads)

[Compare service tiers](#)

DTUs

[Compare DTU options](#)

5 (Basic)

Data max size (GB)



Cost summary

Basic (Basic)

Cost per DTU (in INR)

81.49

DTUs selected

x 5

ESTIMATED COST / MONTH

407.46 INR

Apply

• Choose **Locally-redundant backup storage**

Home > Azure SQL | SQL databases

Create SQL Database ...

Microsoft

Want to use SQL elastic pool? ⓘ

☐ Yes ☒ No

Workload environment

☒ Development

☐ Production

 Default settings provided for Development workloads. Configurations can be modified as needed.

Compute + storage * ⓘ

Basic

2 GB storage

[Configure database](#)

Backup storage redundancy

Choose how your PITR and LTR backups are replicated. Geo restore or ability to recover from regional outage is only available when geo-redundant storage is selected.

Backup storage redundancy ⓘ

☒ **Locally-redundant backup storage**

☐ Zone-redundant backup storage

☐ Geo-redundant backup storage

☐ Geo-Zone-redundant backup storage

- Keep the default settings if performing the lab.

Step 6: Configure Networking

- Choose the Connectivity method **Public endpoint**.
- Enable **Allow Azure services and resources to access this server** if required.
- **Add your client IP address** to the firewall rules.

- Leave other networking settings as default unless specified.

Microsoft Azure

Home > Azure SQL | SQL databases

Create SQL Database

Microsoft

Basics **Networking** Security Additional settings Tags Review + create

Configure network access and connectivity for your server. The configuration selected below will apply to the selected server 'web-db' and all databases it manages. [Learn more](#)

Network connectivity

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#)

Connectivity method *

☐ No access

☒ **Public endpoint**

☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server * ☐ No ☒ **Yes**

Add current client IP address * ☐ No ☒ **Yes**

Cost summary	
Basic (Basic)	
Cost per DTU (in INR)	81.49
DTUs selected	x 5
ESTIMATED COST / MONTH	407.46 INR

Step 7: Review and Create

- Review all the configuration details.
- Click **Review + Create**.
- Click **Create** to start deployment.

Microsoft Azure

Home > Azure SQL | SQL databases

Create SQL Database

Microsoft

Basics Networking Security Additional settings Tags **Review + create**

Product details

SQL database by Microsoft

[Terms of use](#) | [Privacy policy](#)

Estimated cost per month
407.46 INR

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Subscription	Visual Studio Enterprise Subscription
Resource group	terraform-storage-rg
Region	East Asia
Database name	SQLsvr-dev-eus
Server	(new) web-db
Authentication method	SQL authentication
Server admin login	SQLadmin
Compute + storage	Basic: 2 GB storage
Backup storage redundancy	Locally-redundant backup storage

Create < Previous [Download a template for automation](#)

Cost summary	
Basic (Basic)	
Cost per DTU (in INR)	81.49
DTUs selected	x 5
ESTIMATED COST / MONTH	407.46 INR

Step 8: Wait for Deployment

- Wait until the SQL Database deployment completes.
- Click **Go to Resource** after deployment is successful.

Microsoft Azure

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Copilot

Home

Microsoft.SQLDatabase.newDatabaseNewServer_7fe5516384424a219e9b2 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_7fe5... Start time : 7/14/2026, 9:46:28 AM

Subscription : Visual Studio Enterprise Subscription Correlation ID : 0630daa4-8494-4f66-b956-8d4322faf0b7

Resource group : terraform-storage-rg

> Deployment details

> Next steps

Go to resource

Step 9: Connect to the Database

- Open the SQL Database Overview page.
- Copy the Server name if required.
- Connect using Azure Query Editor
- Sign in using the SQL Server Administrator credentials.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Home > Microsoft.SQLDatabase.newDatabaseNewServer_7fe5516384424a219e9b2 | Overview > SQLsvr-dev-eus (web-db/SQLsvr-dev-eus)

SQLsvr-dev-eus (web-db/SQLsvr-dev-eus) | Query editor (preview)

SQL database

Search

Feedback Classic experience

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Resource visualizer

Settings

Data management

Integrations

Power Platform

Security

Intelligent performance

Monitoring

Automation

Help

Query Editor

Run queries on your database directly in the browser

Connecting to SQLsvr-dev-eus on web-db

Microsoft Entra authentication

SQL authentication

Username *

SQLadmin

Password *

Enter password

Connect

Forgot password? Reset in logical server settings

Step 10: Verify the Database

- Open the Query Editor.
- Create a sample table if required.
- Insert a few sample records.

CREATE TABLE Employees

```
(  
  Id INT IDENTITY(1,1) PRIMARY KEY,  
  Name NVARCHAR(100) NOT NULL,  
  Email NVARCHAR(100) NOT NULL,  
  Department NVARCHAR(100) NOT NULL,  
)
```

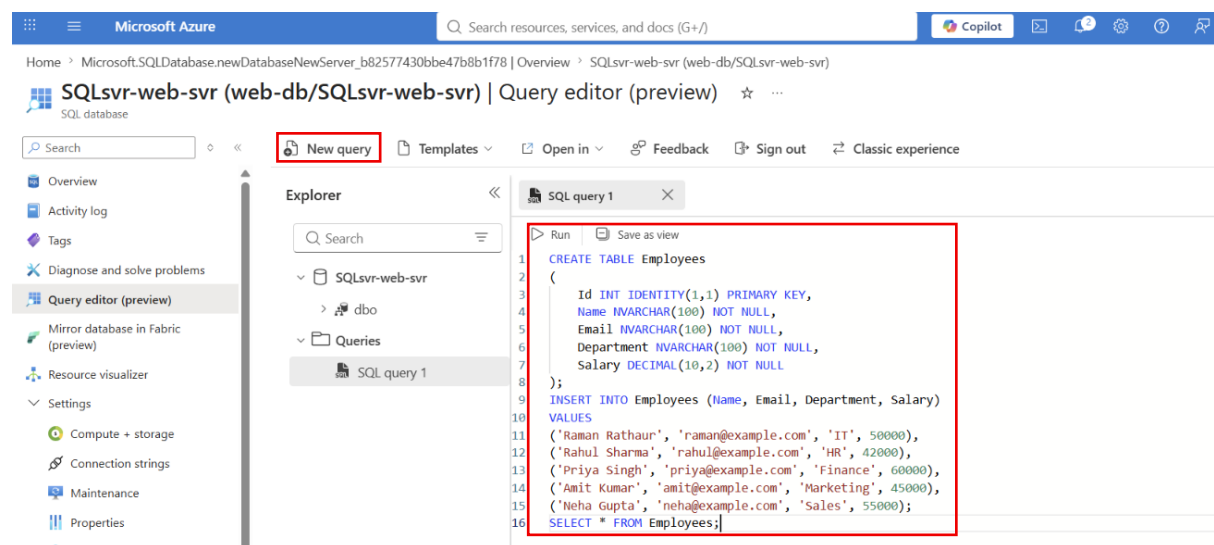
```

Salary DECIMAL(10,2) NOT NULL
);
INSERT INTO Employees (Name, Email, Department, Salary)
VALUES
('Raman Rathaur', 'raman@example.com', 'IT', 50000),
('Rahul Sharma', 'rahul@example.com', 'HR', 42000),
('Priya Singh', 'priya@example.com', 'Finance', 60000),
('Amit Kumar', 'amit@example.com', 'Marketing', 45000),
('Neha Gupta', 'neha@example.com', 'Sales', 55000);
SELECT * FROM Employees;

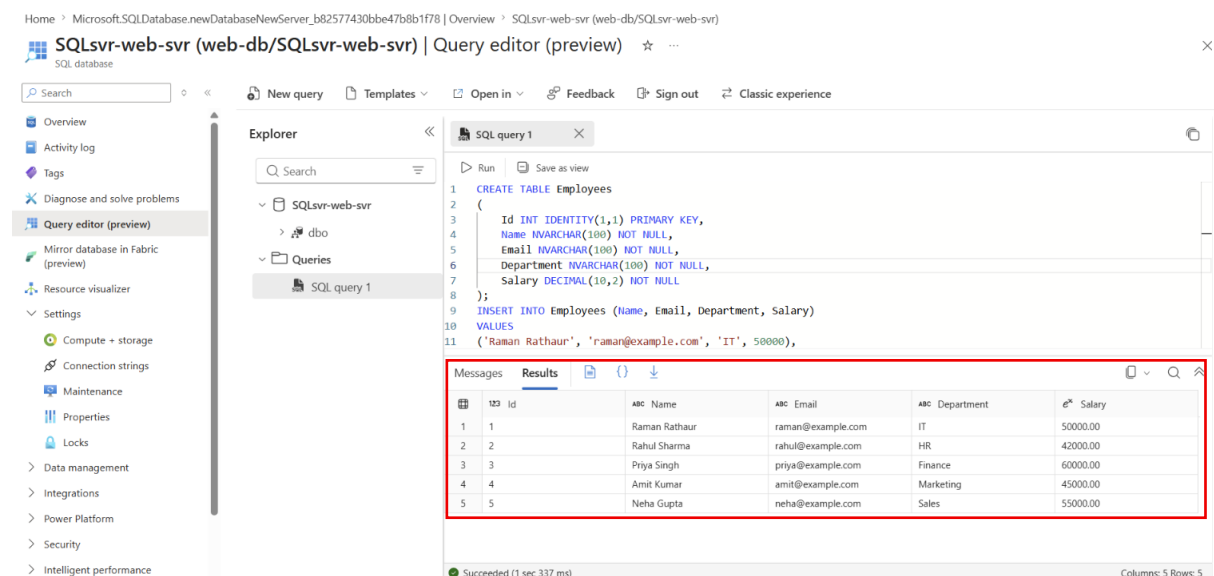
```

3. View All Records

- Click on the Run



- Run a **SELECT** query to verify the data.
- SELECT * FROM Employeee;**



Employee List

Id	Name	Email	Department	Salary
5	Neha Gupta	neha@example.com	Sales	55000.00
4	Amit Kumar	amit@example.com	Marketing	45000.00
3	Priya Singh	priya@example.com	Finance	60000.00
2	Rahul Sharma	rahul@example.com	HR	42000.00
1	Raman Rathaur	raman@example.com	IT	50000.00